RAPPORT : Architecture MVC et Déploiement Cloud (Atelier 7)

- **1. Objectif de l'Atelier**

  Mettre en place une architecture **MVC (Modèle-Vue-Contrôleur)** pour gérer dynamiquement les produits d'une boutique en ligne. L'objectif final est de permettre le filtrage des produits par catégorie et de déployer la solution sur **Vercel** avec une base de données distante **AlwaysData**.

- **2. Modification de la Base de Données (Migration)**

  Action : Ajout de la colonne category à la table products pour permettre le tri (ex: "Pottery", "Food").

  - **Fichier :** create_products_table.php

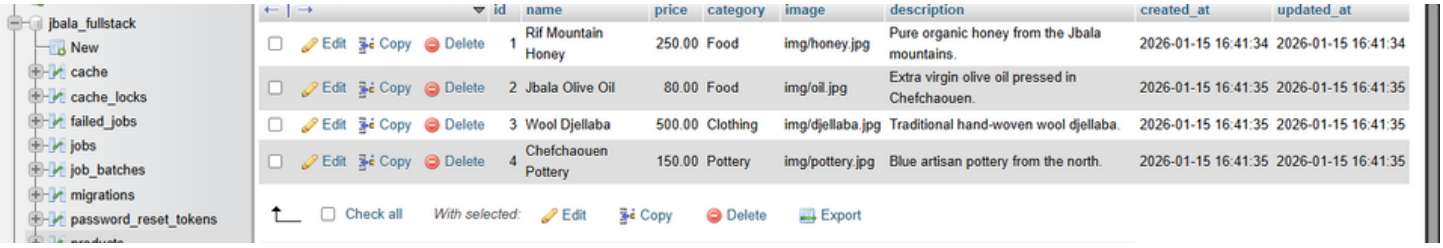
```php
return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create(table: 'products', callback: function (Blueprint $table): void {
            $table->id();
            $table->string(column: 'name');
            $table->decimal(column: 'price', total: 8, places: 2);
            $table->string(column: 'category');      // Important for Atelier 7
            $table->string(column: 'image');
            $table->text(column: 'description')->nullable();
            $table->timestamps();
        });
    }

    /**
```

- **3. Insertion des Données (Seeding)**

  Action : Création d'un jeu de données réaliste pour "Jbala Peak" (Tagine, Djellaba, Miel) avec leurs catégories respectives.

  - **Fichier :** ProductSeeder.php
  - **Commande :** php artisan db:seed --class=ProductSeeder

```php
public function run(): void
{
    Product::create(attributes: [
        'name' => 'Rif Mountain Honey',
        'price' => 250.00,
        'category' => 'Food',
        'image' => 'img/honey.jpg',
        'description' => 'Pure organic honey from the Jbala mountains.'
    ]);

    Product::create(attributes: [
        'name' => 'Jbala Olive Oil',
        'price' => 80.00,
        'category' => 'Food',
        'image' => 'img/oil.jpg',
        'description' => 'Extra virgin olive oil pressed in Chefchaouen.'
    ]);

    Product::create(attributes: [
        'name' => 'Wool Djellaba',
        'price' => 500.00,
        'category' => 'Clothing',
        'image' => 'img/djellaba.jpg',
        'description' => 'Traditional hand-woven wool djellaba.'
    ]);
```



id	name	price	category	image	description	created_at	updated_at
1	Rif Mountain Honey	250.00	Food	img/honey.jpg	Pure organic honey from the Jbala mountains.	2026-01-15 16:41:34	2026-01-15 16:41:34
2	Jbala Olive Oil	80.00	Food	img/oil.jpg	Extra virgin olive oil pressed in Chefchaouen.	2026-01-15 16:41:35	2026-01-15 16:41:35
3	Wool Djellaba	500.00	Clothing	img/djellaba.jpg	Traditional hand-woven wool djellaba.	2026-01-15 16:41:35	2026-01-15 16:41:35
4	Chefchaouen Pottery	150.00	Pottery	img/pottery.jpg	Blue artisan pottery from the north.	2026-01-15 16:41:35	2026-01-15 16:41:35

- **4. La Logique Métier (Le Contrôleur)**

  Action : Création du ProductController pour gérer deux fonctionnalités :

  1. **index()** : Afficher tous les produits.
  2. **filter($category)** : Filtrer les produits selon la catégorie demandée via l'URL.

```php
<?php

namespace App\Http\Controllers;

use Illuminate\Http\Request;
use App\Models\Product;

class ProductController extends Controller
{
    public function index(): View
    {
        $products = Product::all();
        return view(view: 'products', data: ['products' => $products]);
    }

    public function filter($category): View
    {
        $products = Product::where(column: 'category', operator: $category)->get();

        return view(view: 'products', data: [
            'products' => $products,
            'currentCategory' => $category
        ]);
    }
}
```

- **5. Le Routage et la Vue**

  Action : Configuration des routes dynamiques et création de la vue products.blade.php pour l'affichage.

  - **Route :** Route::get('/category/{cat}', ...)

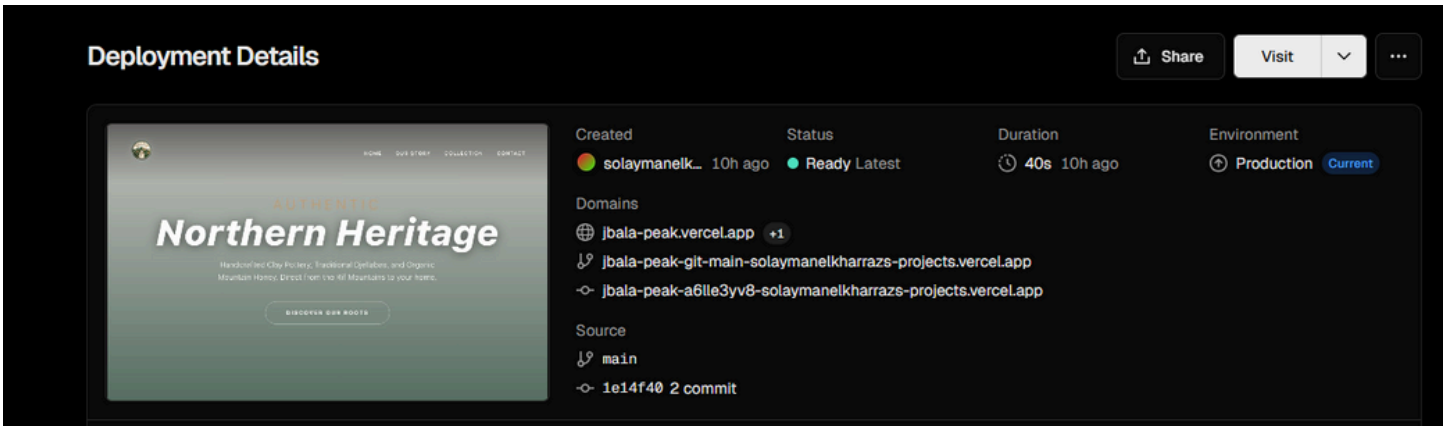
laravel > jbalapeak > routes > web.php > ...

```php
<?php

use Illuminate\Support\Facades\Route;
use App\Http\Controllers\ProductController;

Route::get(uri: '/', action: function (): View {
    $products = \App\Models\Product::take(3)->get();
    return view(view: 'welcome', data: ['products' => $products]);
})->name(name: 'home');

Route::get(uri: '/products', action: [ProductController::class, 'index'])->name(name: 'shop.index');

Route::get(uri: '/category/{cat}', action: [ProductController::class, 'filter'])->name(name: 'shop.filter');

Route::get(uri: '/about', action: function (): View { return view(view: 'about'); });
Route::get(uri: '/contact', action: function (): View { return view(view: 'contact'); });
```

```
laravel > jbalapeak > resources > views > products.blade.php > html
2   <html lang="en">
57  <body>
73      <div class="page_header">
80          @endif
81          <div class="divider"></div>
82      </div>
83
84      <section class="section">
85          <div class="cards-container">
86              @foreach($products as $product)
87              <div class="product-card">
88                  <div class="product-image-box">
89                      <img src="{{ asset(path: $product->image) }}" alt="{{ $product->name }}">
90                  </div>
91                  <div class="product-details">
92                      <a href="{{ route(name: 'shop.filter', parameters: $product->category) }}" class="categor
93                          {{ $product->category }}
94                      </a>
95
96                      <h3 class="product-title">{{ $product->name }}</h3>
97                      <span class="product-price">{{ $product->price }} MAD</span>
98                      <a href="#" class="btn-add-cart">Add to Cart</a>
99                  </div>
100             </div>
101             @endforeach
102         </div>
103     </section>
104
105     <footer>
106         <p>&copy; 2026 Jbala Peak. All Rights Reserved.</p>
107     </footer>
```

- **6. Déploiement (Vercel & AlwaysData)**

  Action : Configuration des variables d'environnement sur Vercel (DB_HOST, APP_KEY, etc.) pour connecter l'application "Serverless" à la base de données MySQL distante.



- **7. Liens du Projet**
  - **Code Source (GitHub) :** https://github.com/solaymanelkharraz/Jbala_Peak
  - **Site en Ligne (Vercel) :** https://jbala-peak.vercel.app